

X

<https://swayam.gov.in>https://swayam.gov.in/nc_details/NPTEL

abhishek94056@gmail.com ▾

NPTEL (<https://swayam.gov.in/explorer?ncCode=NPTEL>) » Introduction to Database Systems (course)

Course outline

[About NPTEL \(\)](#)[Introduction to Database Systems \(\)](#)[Week 0 \(\)](#)[Week 1 \(\)](#)[Week 2 \(\)](#)[Week 3 \(\)](#)

Week 4 : Assignment 4

The due date for submitting this assignment has passed.

Due on 2026-02-18, 23:59 IST.

Assignment submitted on 2026-02-05, 18:31 IST

1 point

1) Consider the statements given below:

S1: Every TRC query is safe.

S2: There can be more than one free variable in a condition of a TRC query.

Choose the correct option:

☐ S1: True; S2: True

☐ S1: True; S2: False

Week 4 ()

- ☐ Example Queries in Relation Model and Outer Join Operation (unit? unit=39&lesson=40)
- ☐ Convert ER-Model to a Relational Model (unit? unit=39&lesson=41)
- ☐ Introduction to tuple relational calculus (unit? unit=39&lesson=42)
- ☐ Example TRC queries (unit? unit=39&lesson=43)
- ☐ Lecture Slides (unit? unit=39&lesson=44)
- ☐ Week 4 Feedback Form: Introduction to Database Systems!! (unit? unit=39&lesson=45)

● Quiz: Week 4 : Assignment 4 (assessment? name=193)

Week 5 ()**Week 6 ()**

- ☐ S1: False; S2: False
- ☒ S1: False; S2: True

Yes, the answer is correct.

Score: 1

Accepted Answers:

S1: False; S2: True

2) Consider the sets S1 and S2 below. S1 has different types of keys and S2 has possible constraints on the number of such keys in a relation. Choose the correct matching: **1 point**

S1: {w: primary key; x: candidate key; y: super key; z: foreign key}

S2: { p: zero or more, q: exactly one, r: one or more }

Choose the correct mapping:

- ☐ w--p; x--p; y--q; z--r
- ☐ w--q; x--q; y--p; z--r
- ☒ w--q; x--r; y--r; z--p
- ☐ w--r; x--r; y--p; z--q

Yes, the answer is correct.

Score: 1

Accepted Answers:

w--q; x--r; y--r; z--p

3) Consider the statements given below: **1 point**

S1: If the *candidate key* consists of n attributes for a relation R, then *primary key* attribute-set always has less than n attributes for the same relation.

S2: If a relation does not have a *candidate key* still it can have a *primary key* .

Choose the correct option:

- ☐ S1: True; S2: True

[Week 7 \(\)](#)
[Week 8 \(\)](#)
[Week 9 \(\)](#)
[Week 10 \(\)](#)
[Week 11 \(\)](#)
[Week 12 \(\)](#)
[Download Videos \(\)](#)
[Lecture Notes \(\)](#)
[Problem Solving
Session - Jan 2026 \(\)](#)

- ☐ S1: True; S2: False
☒ S1: False; S2: False
☐ S1: False; S2: True

Yes, the answer is correct.

Score: 1

Accepted Answers:

S1: False; S2: False

4) Consider the sets mentioned in the table below:

1 point

Set S1	Set S2
w: Domain constraint	p: Tuples in one relation do not refer to non-existing tuples in other relations.
x: Key constraint	q: No two tuples in the relation have the same values for any key
y: Entity-Integrity constraint	r: Primary-key attributes do not contain NULL values.
z: Referential-Integrity constraint	s: Every domain must contain atomic values

Choose the correct matching:

- ☐ w--s; x--q ; y--r ; z--p
☐ w--s; x--p ; y--p; z--r
☒ w--q; x--r ; y--r; z--p
☐ w--q; x-- s; y--p; z--r

No, the answer is incorrect.

Score: 0

Accepted Answers:

w--s; x--q ; y--r ; z--p

5) For a relational algebra operator θ , we can rewrite $r_1 \theta r_2$ by using other relational algebra operators appropriately. Choose the **incorrect** option:

1 point

- ☐ *Intersection* : the re-written expression requires only $\cup$ and $-$

☐

Join : the re-written expression requires only $\times$ and σ .

☒

Join : the re-written expression requires only Π and σ .

☐

Division : the re-written expression requires only Π , $\times$, and $-$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Join : the re-written expression requires only Π and σ .

6) In the tuple relational calculus (TRC) query, one can not give ____ of how to evaluate a query. Therefore, we write a **1 point** ____ expression to specify the data retrieval requirement. Fill the blanks:

☐

Data; Procedural

☐

Description; Procedural

☒

Data; Declarative

☐

Description; Declarative

No, the answer is incorrect.

Score: 0

Accepted Answers:

Description; Declarative

7) Consider the TRC query given below:

1 point

$\{e.name \mid employee(e) \wedge e.sex = 'M' \wedge (\exists d)(department(d) \wedge d.name = 'Finance' \wedge e.dept = d.deptId)\}$

In the above TRC query, e is a ____ tuple variable whereas d is a ____ tuple variable. Fill in the blanks:

☐

Free; Free

☒

Free; Bound

☐

Bound; Free

☐

Bound; Bound

Yes, the answer is correct.

Score: 1

Accepted Answers:

Free; Bound

8) Consider the given relational schemas:

1 pointProfessor(pid , pname, age, sex, deptNo, yearOfJoining)Department(did , dname, hod, phoneNo)

Choose the appropriate TRC query to retrieve the department names that do not have any female professors:

- ☒ $\{d. dname | Department(d) \wedge \sim (\exists(p)(Professor(p) \wedge p. sex = 'F' \wedge p. deptNo = d. did))\}$
- ☐ $\{d. dname | Department(d) \wedge (\forall(p)(Professor(p) \wedge p. sex \neq 'F' \wedge p. deptNo = d. did))\}$
- ☐ $\{d. dname | Department(d) \wedge \sim (\forall(p)(Professor(p) \wedge p. sex = 'F' \wedge p. deptNo = d. did))\}$
- ☐ $\{d. dname | Department(d) \wedge \sim (\exists(p)(Professor(p) \wedge p. sex \neq 'F' \wedge p. deptNo = d. did))\}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

 $\{d. dname | Department(d) \wedge \sim (\exists(p)(Professor(p) \wedge p. sex = 'F' \wedge p. deptNo = d. did))\}$

9) Consider the following schema:

1 pointStudent(rollNo , name, degree, year, sex, deptNo, advisor)Courses(courseId , cname, credits, deptNo)Enrollment(rollNo, courseId, sem, year , grade)Consider the following data retrieval request "Find the *rollNo* of students who have enrolled for exactly one course", and the TRC query for the data retrieval request is given below:

$$\{s. rollNo | Student(s) \wedge \exists e_1 (Enrollment(e_1) \wedge (e_1. rollNo = s. rollNo)) \wedge \\ \neg((\exists e_2)(\exists e_3)(Enrollment(e_2) \wedge Enrollment(e_3) \wedge e_2. rollNo \text{ op}_1 e_3. rollNo \wedge \\ e_2. courseId \text{ op}_2 e_3. courseId \wedge s. rollNo = e_2. rollNo)))\}$$

Choose the correct operators for placeholders *op*₁ and *op*₂ :

- ☒ = and $\neq$
- ☐ = and =
- ☐ $\neq$ and $\neq$
- ☐ $\neq$ and =

Yes, the answer is correct.

Score: 1

Accepted Answers:

= and $\neq$

10) Consider the following schema:

1 point

Suppliers(sid , sname, address)

Parts(pid , pname, color)

Catalog(sid, pid , cost)

and the relational algebra query given below:

$$\pi_{sname}(\pi_{sid}(\pi_{pid}(\sigma_{color='red'}(Parts)) * Catalog) * Suppliers)$$

Choose the equivalent TRC query:

- ☒ $\{S.sname | Suppliers(S) \wedge \exists P \exists C (Parts(P) \wedge P.color = 'red' \wedge Catalog(C) \wedge C.pid = P.pid \wedge C.sid = S.sid)\}$
- ☐ $\{S.sname | Suppliers(S) \wedge \exists P \forall C (Parts(P) \wedge P.color = 'red' \wedge Catalog(C) \wedge C.pid = P.pid \wedge C.sid = S.sid)\}$
- ☐ $\{S.sname | Suppliers(S) \wedge \forall P \exists C (Parts(P) \wedge P.color = 'red' \wedge Catalog(C) \wedge C.pid = P.pid \wedge C.sid = S.sid)\}$
- ☐ $\{S.sname | Suppliers(S) \wedge \forall P \forall C (Parts(P) \wedge P.color = 'red' \wedge Catalog(C) \wedge C.pid = P.pid \wedge C.sid = S.sid)\}$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\{S.sname | Suppliers(S) \wedge \exists P \exists C (Parts(P) \wedge P.color = 'red' \wedge Catalog(C) \wedge C.pid = P.pid \wedge C.sid = S.sid)\}$

